- Overloading and late static binding
- Interfaces, abstract classes, the `final` keyword
- Inheritance and interfaces
- Traits

As OOP evolved over time, so too did the ideas on how best to use it. Objects on their own did not make software cheaper to produce or less error-prone. A milestone paper on the subject of object-oriented design (OOD) was published in 1982, the appropriately titled *Object-Oriented Analysis and Design* by Grady Booch (and was quite possibly the first publication to use the phrase). Booch proposed a design methodology that "controls the complexity of the problem solution" by "decomposing a system into modules."

Booch's methodology began with defining the problem, then the objects and attributes pertinent to that problem, followed by operations on the objects and the interfaces required. Only after this has been done is any implementation considered. We've covered a great deal of the implementation details of PHP's object model. Here is where we will learn how to make the best use of it.

As stated, this chapter expands on the foundational ideas introduced in Chapter 6, "Object Fundamentals." Here we will focus less on what the PHP object model is and more on why it has the features that it does. That chapter was all about explaining how objects in PHP work and the language tools that exist to facilitate OOP. This chapter is concerned with how to write good (or at least pain-free) OOP code, and we're going to focus on object-oriented analysis, designing classes, some fundamental principles of good OOP, and design patterns.

Analysis and Design

Before writing a single line of code, it is necessary to know what the software needs to do, and for that, one must first understand the problem that needs to be solved. We don't create software for its own sake. It is always intended to be used to address the needs of the people that it will end up serving. A little forethought before diving in is always a good idea. Like the saying goes, weeks of programming can save you days of planning! (In other words, time devoted to planning is time saved.)